

MLOps

Explainable AI – CI/CD – Responsible AI

Session 9

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Explainable AI

Motivation

- Many of the state-of-the-art machine learning models are functionally ‘**black boxes**’, as it is nearly impossible to get a feeling for its inner workings.
- This brings us to a question of trust: do I trust that a certain prediction from the model is correct? Or do I even trust that the model is making reasonable predictions in general?
- We want models to be **not only good** in terms of predictive capabilities, **but also interpretable**



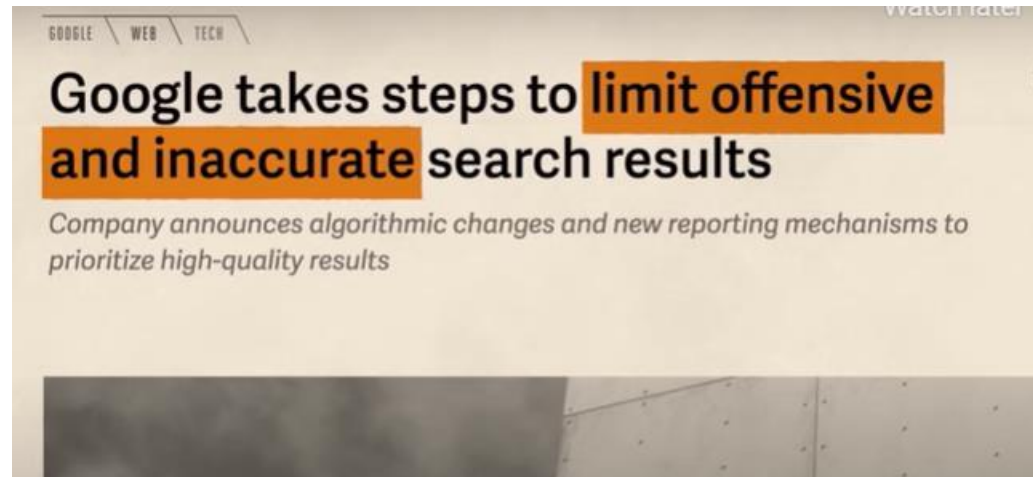
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When is interpretability needed?

1. Identify and mitigate bias
2. Accounting for the context of the problem
3. Improving generalization and performance
4. Ethical and legal reasons

1. Identify and Mitigate Bias

- Bias is potentially present in any dataset, and it is up to the data scientist to identify and attempt to fix it
- Datasets can be limited in size, and they might not be representable for the full population, or the data capturing process might have not accounted for potential biases
- Biases often only become apparent after thorough data analysis or when the relation between model predictions and the model input is analyzed



2. Accounting for the Context of the Problem

- In most problems, you are working with a dataset that is only a rough representation of the problem you are trying to solve, and a machine learning model can typically not capture the full complexity of the real-life task
- An interpretable model helps you to **understand and account for the factors** that are (not) included in the model and account for the context of the problem when taking actions based on model predictions.

3. Improving Generalization and Performance

- A high interpretability typically leads to a model that generalizes better
- Interpretability is not about understanding every single detail of the model for all of the data points
- The combination of solid data, model and problem understanding is necessary to have a solution that performs better.

4. Ethical and Legal Reasons

- In industries like finance and healthcare it is essential to audit the decision process and ensure it is e.g., not discriminatory or violating any laws
- With the rise of data and privacy protection regulation like GDPR, interpretability becomes even more essential
- There are applications (medical applications or self-driving cars) where a single incorrect prediction can have a significant impact. and being able to 'verify' the model is critical. Therefore, the system should be able to explain how it reached a given recommendation.

Properties of Interpretable Models

- **Transparency**

- How exactly does the model work?

Simulatability

A human should be able to take input data and model parameters and calculate prediction

Decomposability

Understanding each input, parameter, calculation
Inputs must be interpretable
Models with highly engineered or anonymous features are not decomposable

Algorithmic Transparency

Modern deep learning methods lack this kind of transparency

- **Post-hoc explanations:**

- What else can the model tell me?

Text Explanations

Use words to explain the model -> feature importance

Visualizations

Present different visualizations of inputs in relation to model outcomes

Example explanations

Try to find representative examples

Local Explanations

Explain the model in a small region -> LIME

Why are explanations important?

People that have access to explanations:

Trust: how can we trust the predictions are correct?

Explanations allow to interpret and compare classifiers

Predict: how model will behave on unseen data?

Explanations have helped to better understanding of model behavior

Improve: is it possible to improve the model?

Explanations have helped even non-ML experts to better feature selection

Does this work in all cases?

- Do not blindly embrace post-hoc explanations!
- Can take some time to train surrogate model
- Model may not be locally linear
- Interpretable feature may not be useful predictor
- Too many important features

LIME – SHAP Values

CI / CD

ML Systems differ from other Software Systems

- **Team skills:**

- In an ML project, the team usually includes data scientists or ML researchers, who focus on exploratory data analysis, model development, and experimentation. These members **might not be experienced software engineers** who can build production-class services.

- **Development:**

- ML is experimental in nature. You should try different features, algorithms, modeling techniques, and parameter configurations to find what works best for the problem as quickly as possible. The challenge is **tracking what worked and what didn't, and maintaining reproducibility while maximizing code reusability**.

- **Testing:**

- Testing an ML system is more involved than testing other software systems. In addition to typical unit and integration tests, you need **data validation, trained model quality evaluation, and model validation**.

- **Deployment:**

- In ML systems, deployment isn't as simple as deploying an offline-trained ML model as a prediction service. ML systems can require you to deploy a **multi-step pipeline to automatically retrain and deploy model**. This pipeline adds complexity and requires you to automate steps that are manually done before deployment by data scientists to train and validate new models.

- **Production:**

- ML models can have reduced performance not only due to suboptimal coding, but also due to constantly evolving data profiles. In other words, models **can decay in more ways than conventional software systems**, and you need to consider this degradation. Therefore, you need to track summary statistics of your data and monitor the online performance of your model to send notifications or roll back when values deviate from your expectations.

Differences in CI/CD



- ML and other software systems are similar in continuous integration of source control, unit testing, integration testing, and continuous delivery of the software module or the package.
- However, in ML, there are a few notable differences:

Continuous Integration

CI is no longer only about testing and validating code and components, but also testing and validating **data, data schemas, and models**.

Continuous Delivery

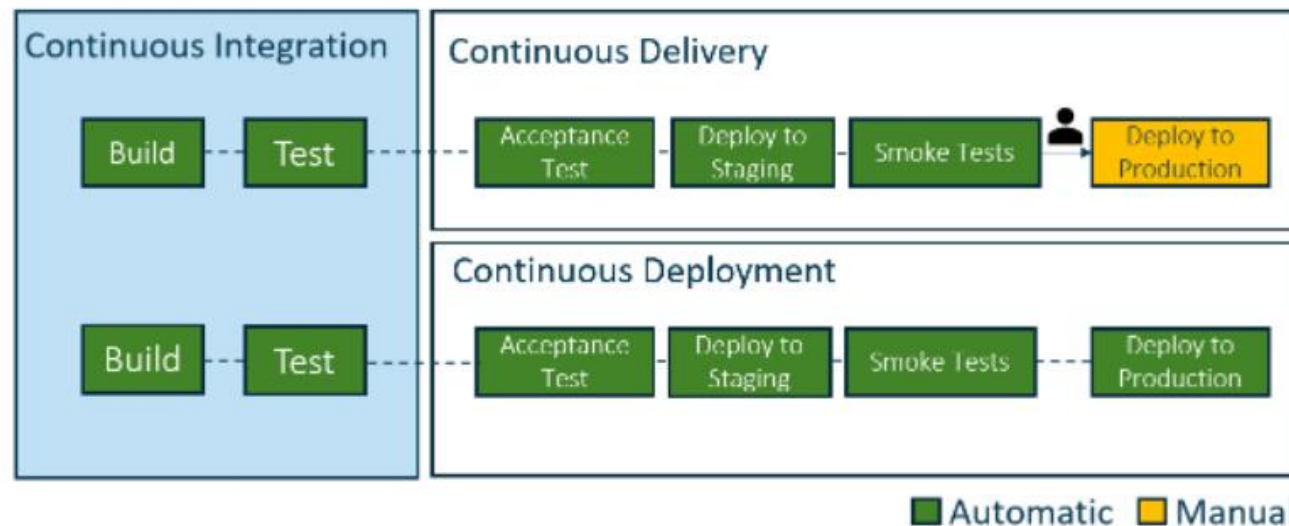
CD is no longer about a single software package or a service, but a system (an **ML training pipeline**) that should automatically deploy another service (model prediction service).

Continuous Training

CT is a new property, unique to ML systems, that's concerned with **automatically retraining and serving the models**.

Continuous integration, delivery, and deployment in MLOps

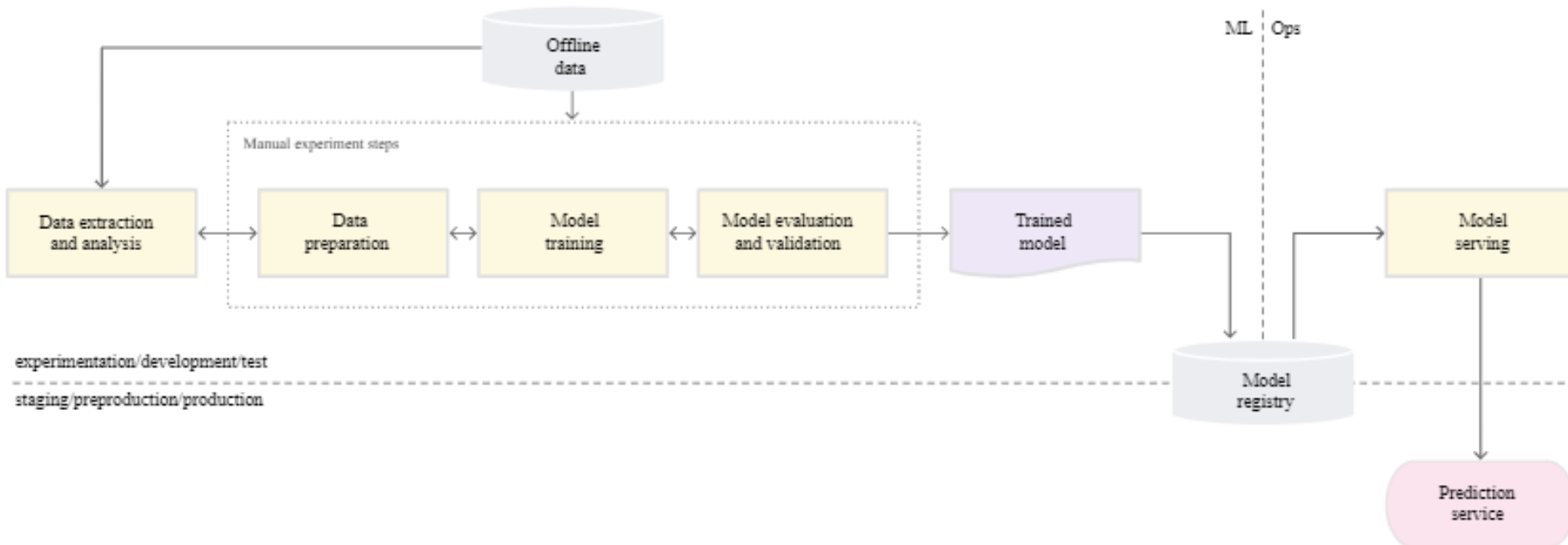
- **Automation** is the primary reason for CI/CD in the MLOps workflow
- The goal of enabling continuous delivery to the ML service is to maintain data and source code versions of the models, enable triggers to perform necessary jobs in parallel, build artifacts, and release deployments for production
- Several cloud vendors are promoting DevOps services to monitor ML services and models in production, as well as orchestrate with other services on the cloud
- Using CI and CD, we can enable continual learning, which is critical for a ML system's success. Without continual learning, a ML system is deemed to end up as a failed Proof of Concept (PoC)



Source: Engineering MLOps: Rapidly build, test, and manage production-ready machine learning life cycles at scale (Emmanuel Raj, 2021)

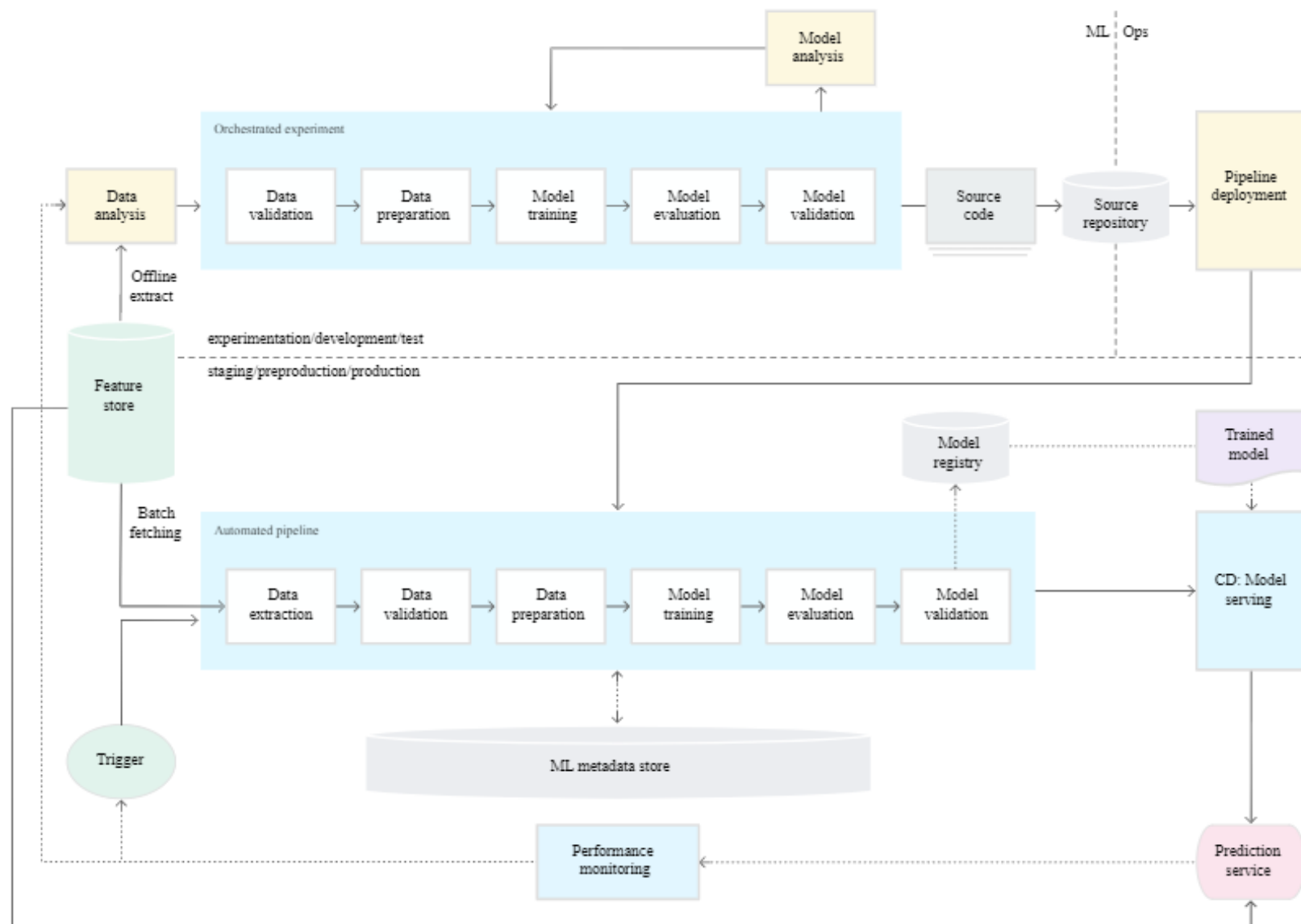
MLOps level 0: Manual Process

- Many teams have data scientists and ML researchers who can build state-of-the-art models, but their process for building and deploying ML models is entirely manual.

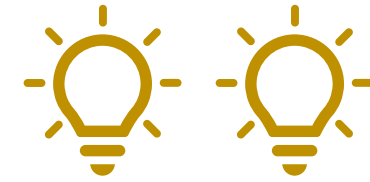


MLOps level 1: ML Pipeline Automation

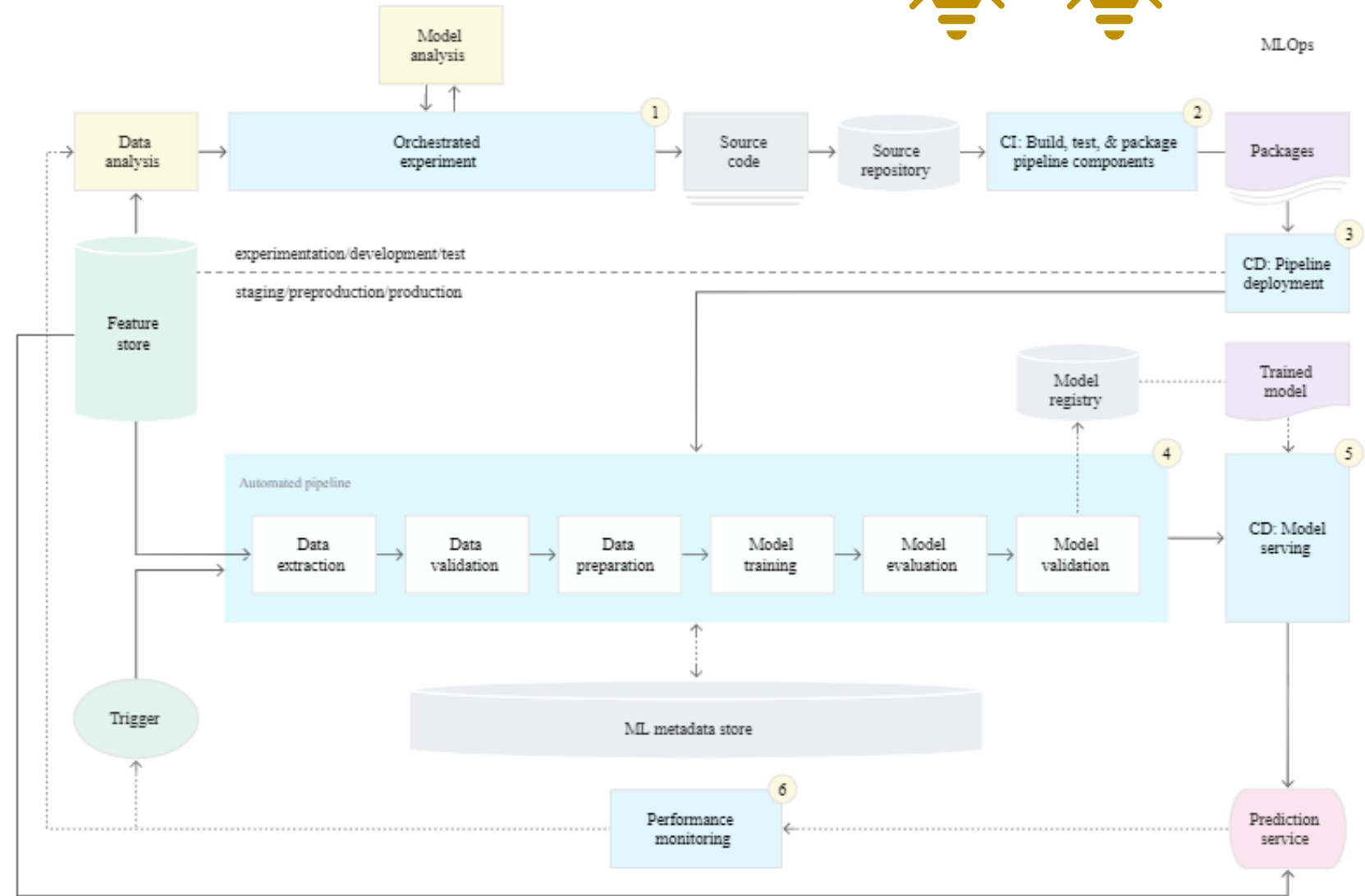
- The goal of level 1 is to perform continuous training of the model by automating the ML pipeline; this lets you achieve continuous delivery of model prediction service.
- To automate the process of using new data to retrain models in production, you need to introduce automated data and model validation steps to the pipeline, as well as pipeline triggers and metadata management.



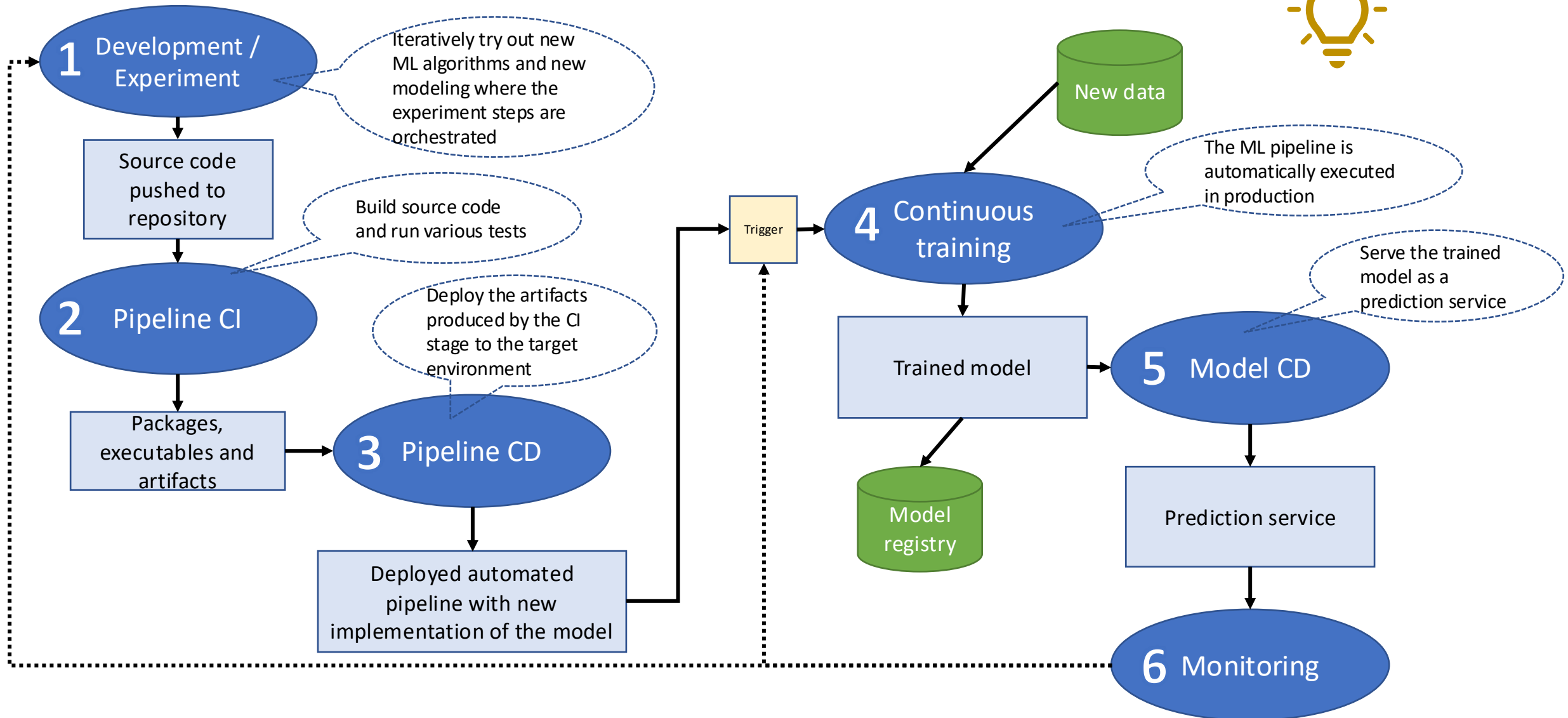
MLOps level 2: CI/CD Pipeline Automation



- For a rapid and reliable update of the pipelines in production, you need a robust automated CI/CD system.
- This automated CI/CD system lets your data scientists rapidly explore new ideas around feature engineering, model architecture, and hyperparameters.
- They can implement these ideas and automatically build, test, and deploy the new pipeline components to the target environment.



Stages of the ML CI/CD Automation Pipeline



Pipeline Execution Triggers

- **Artifactory triggers**

- Artifacts are generated at different stages in the pipeline and development process. Generated artifacts, such as a trained model, metadata, uploaded Docker images, or any file that has been uploaded, can be triggered to execute a certain process in the CI/CD pipeline. Having such options can enable great flexibility and functionality for the CI/CD pipeline.

- **Docker Hub triggers**

- Every time you push a new Docker image to a Docker Hub repository of your choice, a trigger in the CI/CD pipeline can be executed as per requirements. For example, when you upload a new Docker image to Docker Hub (or any cloud Container Registry), the pipeline is triggered to deploy the Docker image as a web service.

- **Schedule triggers**

- Following a specific time schedule. This type of trigger is very useful for a scheduled clean-up or cron jobs or any other workflow that needs to be run following a time interval; for example, a trigger for ML model retraining at 12:00 every day.

- **API triggers**

- The purpose of API triggers is to integrate with external services (or any other application or service you have). This can be set up so your pipeline process is triggered based on an event on another system. For example, when the system admin comments retrain on a developer's platform, the pipeline can be triggered to retrain the existing deployed model. These triggers are facilitated using API calls.

- **Git triggers**

- Git triggers are commonly used to trigger pipeline executions, for instance when new code is committed to a branch or a new pull request is made. When changes are made to a repository, then certain processes can be triggered in the pipeline as per requirements.

Jenkins

Responsible AI

- What is Responsible AI?
- What is Regulatory Compliance?
- How to govern AI?

Responsible AI and Regulatory Compliance

Responsible AI

A responsible use of machine learning (more commonly referred to as Responsible AI) covers two main dimensions:

Intentionality

- Ensuring that models are designed and behave in ways aligned with their purpose.
 - Data used for AI projects comes from compliant and unbiased sources
 - Collaborative approach to AI projects -> multiple checks on potential model bias.
- Intentionality also includes explainability, meaning the results of AI systems should be explainable by humans (ideally, not just the humans that created the system).

Accountability

- Centrally controlling, managing, and auditing the Enterprise AI effort— Accountability is about having an overall view of which teams are using what data, how, and in which models.
- It also includes the need for trust that data is reliable and being collected in accordance with regulations as well as a centralized understanding of which models are used for what business processes. This is closely tied to traceability—if something goes wrong, is it easy to find where in the pipeline it happened?



Establish Responsibilities: RACI Matrix

- RACI stands for **R**esponsible, **A**ccountable, **C**onsulted and **I**nformed
- Highlights the roles of different stakeholders in the overall MLOps process



Tasks	Business stakeholders	Business analysis/citizen DS	Data scientists	Risk/audit	Data ops	Production/exploitation	Resources admin/architect
Identification	A/R	C		I			
Data preparation	C	A/R	C				
Data modeling	C	A	R				
Model acceptance	I	C	C	A/R			
Productionalization		C	A/R	I	C		
Capitalization			R		R		A
Integration to external systems					A/R		
Global orchestration		C			R	A	
User acceptance tests	A/R	R	C		I		
Deployments					R	A	I
Monitoring	I	C				A/R	I

A: accountable

R: responsible

C: consulted

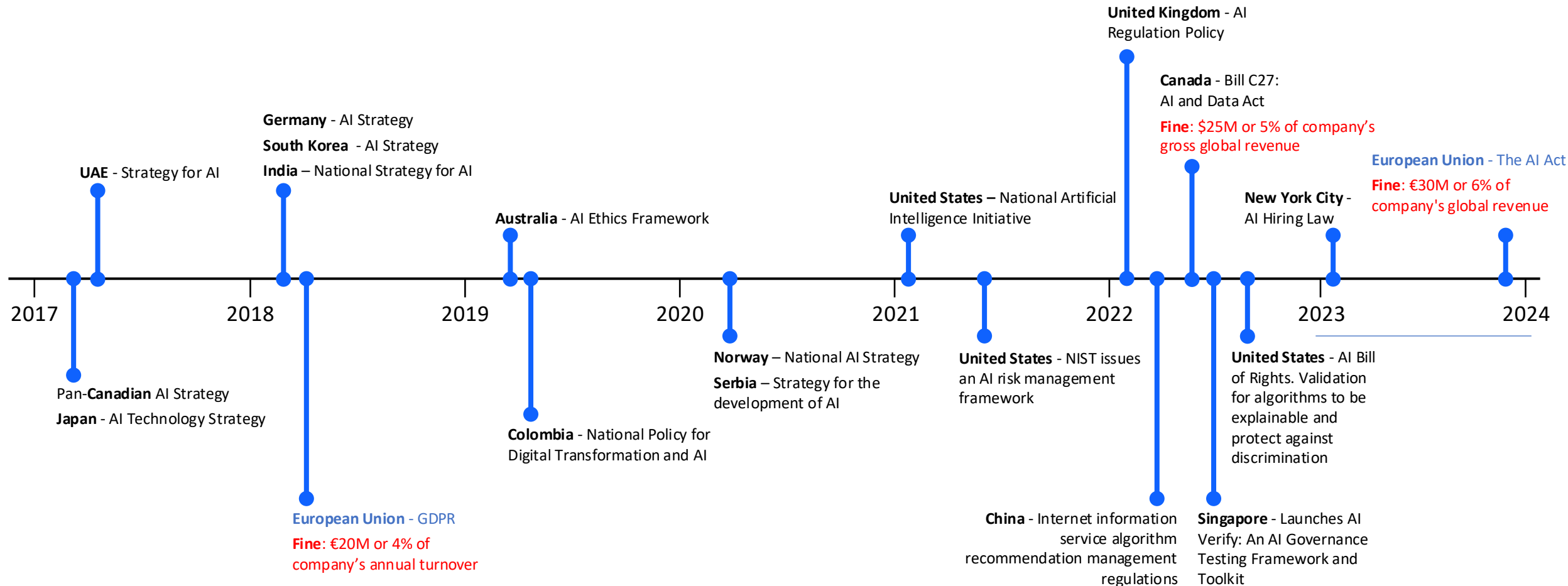
I: informed

Source: Introducing MLOps (Mark Treveil – O'Reilly)

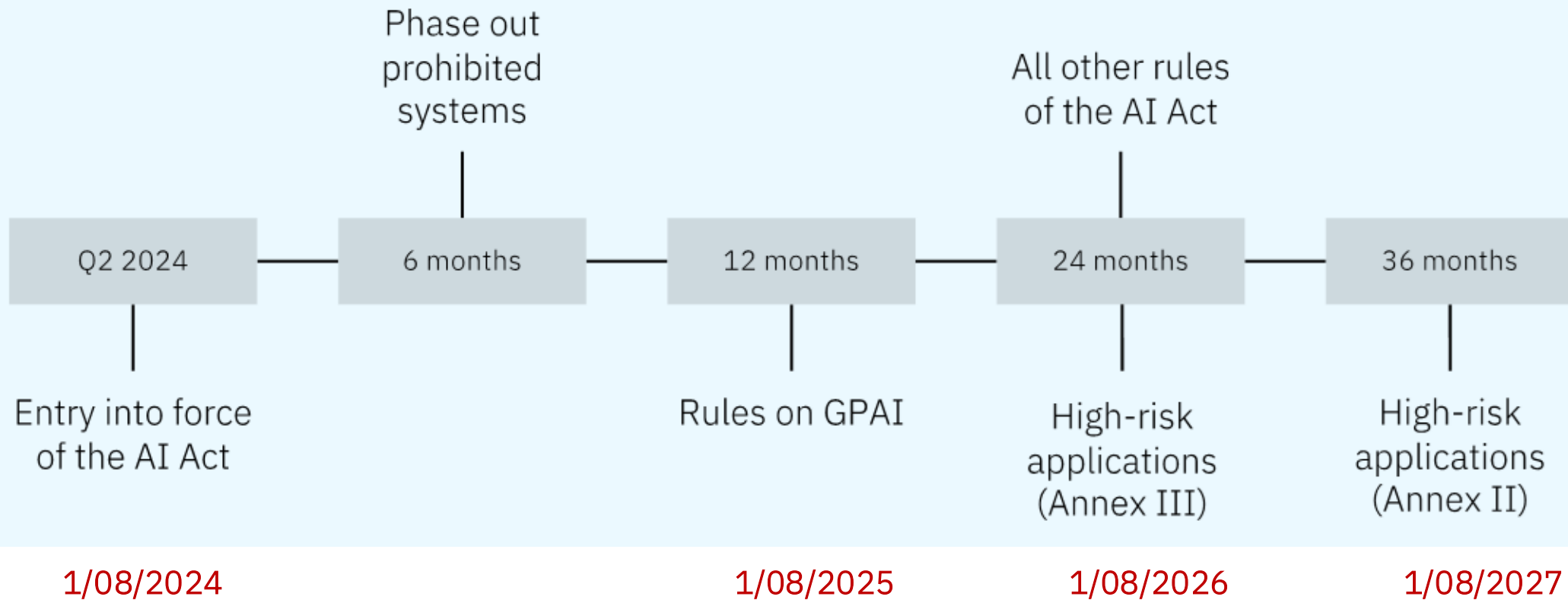
What is regulatory compliance?

- Regulatory compliance is the process of complying with applicable laws, regulations, policies and procedures, standards, and the other rules issued by governments and regulatory bodies
- Violations of regulatory compliance could result in legal punishment, including fines

Regulatory Compliance



EU AI Act Implementation Timeline



EU AI Act Risk Levels

Unacceptable Risk —> Not allowed, so companies need to change their models accordingly. Examples: social scoring systems, real-time remote biometric verification



High Risk —> High-risk applications include services directly affecting citizens' lives. Models will have to be put through strict assessment regimes before they can be put on the market. Apart from this, these high-risk processes require detailed logs and human oversight as well. Examples: credit scoring systems, automated insurance claims, job or educational applications

- Risk Management, metrics and monitoring, record of activities, allow for human supervision
- Transparency to the end user, record and technical documentation

Limited Risk —> Companies need to ensure that they're being completely transparent with their customers about interacting with AI system, what AI is being used for and the data involved. Best practices related to data quality and fairness are essential even in this risk regime. Examples: chatbots, personalization, image and video processing and recommender systems.

Minimal Risk —> Providers are encouraged to have a code of conduct in place that ensures AI is being used ethically, though they are not subject to further regulatory requirements. Examples: applications such as spam filtering or video games

(80-90%)

- Transparency and good practices

<https://www.ibm.com/think/insights/eu-ai-act>

Model Risk Assessment

- Frequently, a set of questions that can help to assess the different elements of risk that are involved in a model deployment

Compliance



Compliance



To adhere to current requirements and prepare for future regulations, Trusted AI programs are used to embed clear guardrails across the organization and continually adapt to address new, evolving risks. AI governance refers to the rules and guidelines that control the development, use, and implementation of AI technologies. It ensures responsible and ethical development of AI technologies that stay in compliance with relevant laws and regulations. The following questions may help to identify the applicable regulatory requirements and risk management strategies based on attributes of your AI System???

default title



Does this use case involve algorithmic decision tools or an artificial intelligence model or system based on machine learning, logic- and knowledge-based approaches, or statistical approaches? *

- ☐ Yes
- ☐ No
- ☐ Not applicable

default title



Is there a possibility to exploit any of the vulnerabilities of a specific group of persons due to their age, physical or mental disability, in order to materially distort the behavior of a person pertaining to that group in a manner that causes or is likely to cause that person or another person physical or psychological harm *

- ☐ Yes
- ☐ No
- ☐ Not applicable

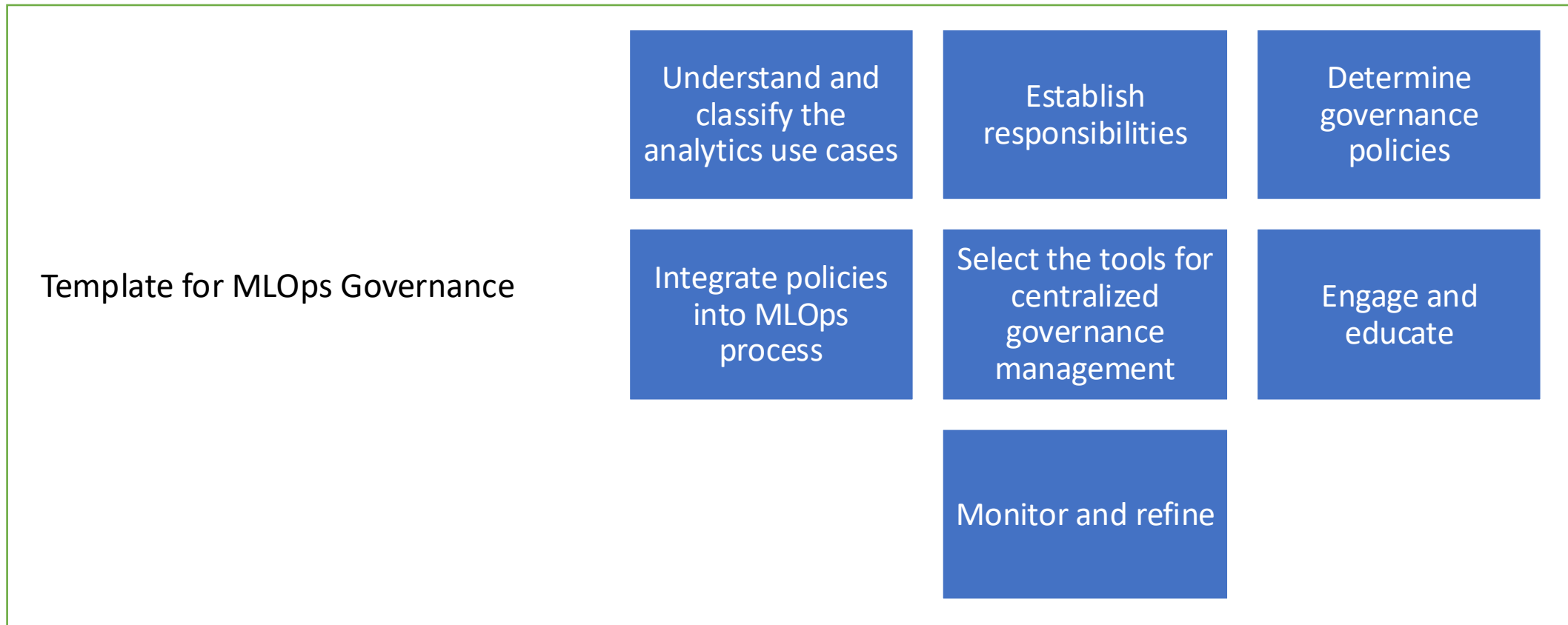
Model Risk Assessment Questionnaire Template

- **Compliance**
 - To adhere to current requirements and prepare for future regulations, Trusted AI programs are used to embed clear guardrails across the organization and continually adapt to address new, evolving risks.
 - Does this use case involve algorithmic decision tools or an artificial intelligence model or system based on machine learning, logic- and knowledge-based approaches, or statistical approaches?
 - Does the AI system follow privacy rights, data protection and security in line with our privacy principles?
- **Objective and proportionality**
 - As with any collection and use of personal data, processing using AI techniques must be carried out with due regard for data protection.
 - Is personal data processed by the AI system?
- **Providers, users of AI systems and individuals**
 - If the data controller is not the AI system provider, the sharing of responsibilities between these two parties must be formalised.
 - These responsibilities must be clear to the individuals involved in the implementation of the processing, for those whose data is processed, as well as those on whom the system will have an impact.
 - Are the legal persons in charge of AI system development, deployment and monitoring clearly defined?
- **Reliability**
 - Could the AI system produce non reliable output?
- **Fairness**
 - Is there potential for discrimination in the AI systems accessibility and/or discrimination?
- **Transparency and Explainability**
 - Can the AI system output be understood and explained?
- **Contestability**
 - Can the user affected challenge or complain about the results?
- **Accountability**
 - Is the AI system outcomes clear and have human oversight?
- **Human Centered Values**
 - Does the AI system account for human rights, diversity and the autonomy of individuals?
- **Human, societal and environmental wellbeing**
 - Can the AI system produce inaccurate outputs (including hallucinations, misinformation, poor advice, manipulation, etc.) that results in loss/ harm?

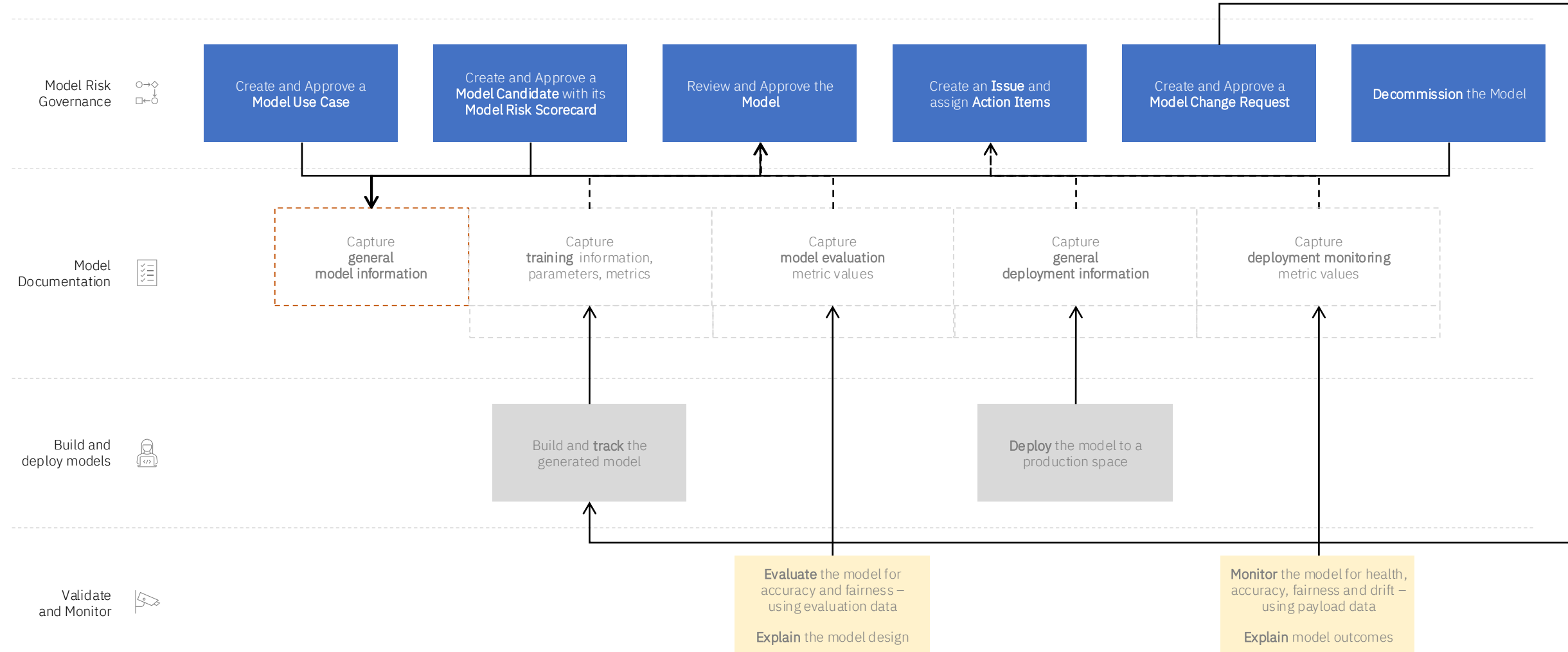
AI Governance

What is MLOps Governance?

- MLOps Governance is the set of controls placed on a business to ensure that it delivers on its responsibilities to all stakeholders, from shareholders and employees to the public and national governments.
- These responsibilities include financial, legal, and ethical obligations. Underpinning all three of these is the fundamental principle of fairness.



Lifecycle Governance flow



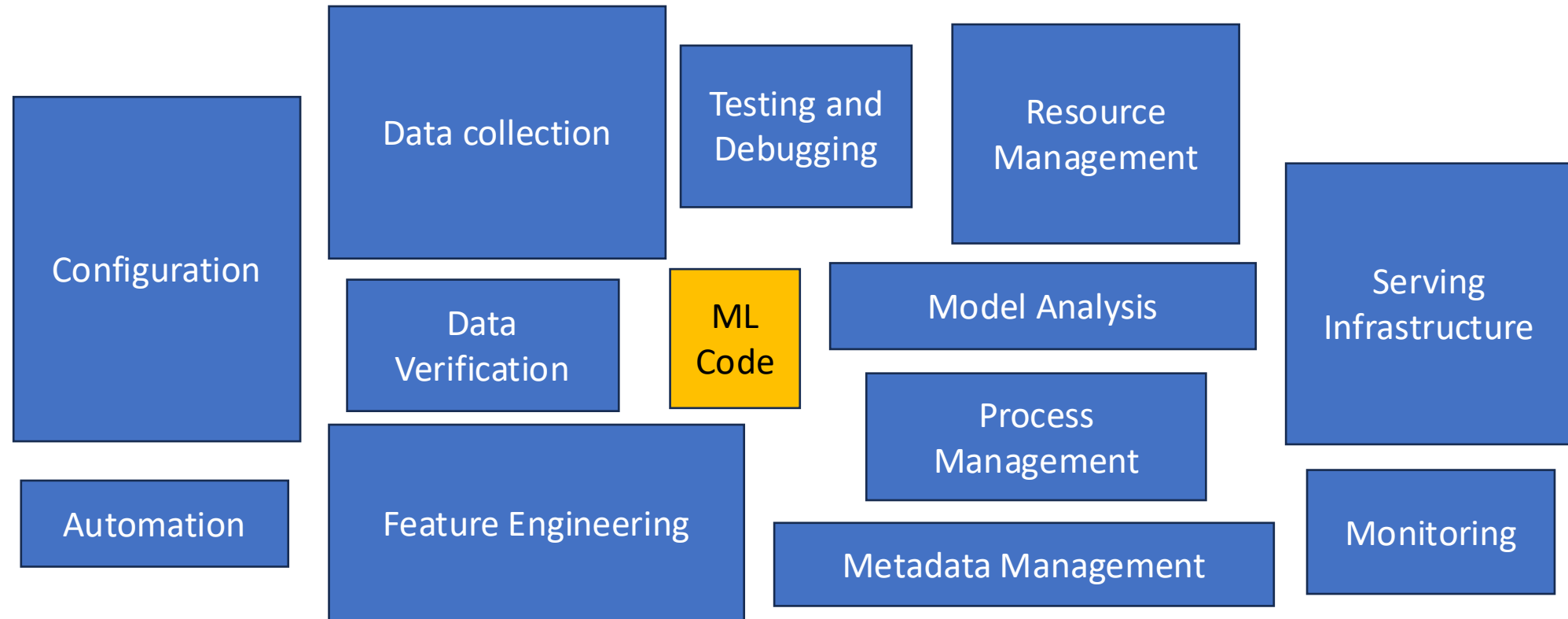
Final Consideration

- It's not possible to successfully manage the model life cycle, mitigate the risks, and deliver value at scale **without governance**.
- **Governance impacts everything** from how the business can acceptably exploit ML, to the data and algorithms that can be used, to the style of operationalization, monitoring, and retraining.
- **MLOps at scale is in its infancy**. Few businesses are doing it, and even fewer are doing it well.
- **Public trust in ML is at risk**. Even without additional legislation, businesses need to worry about the potential damage to their public image that can be caused by an inadvertently harmful model.
- When planning to scale MLOps, **start with governance and use it to drive the process**: It will take time and iteration.

Kubeflow – Airflow

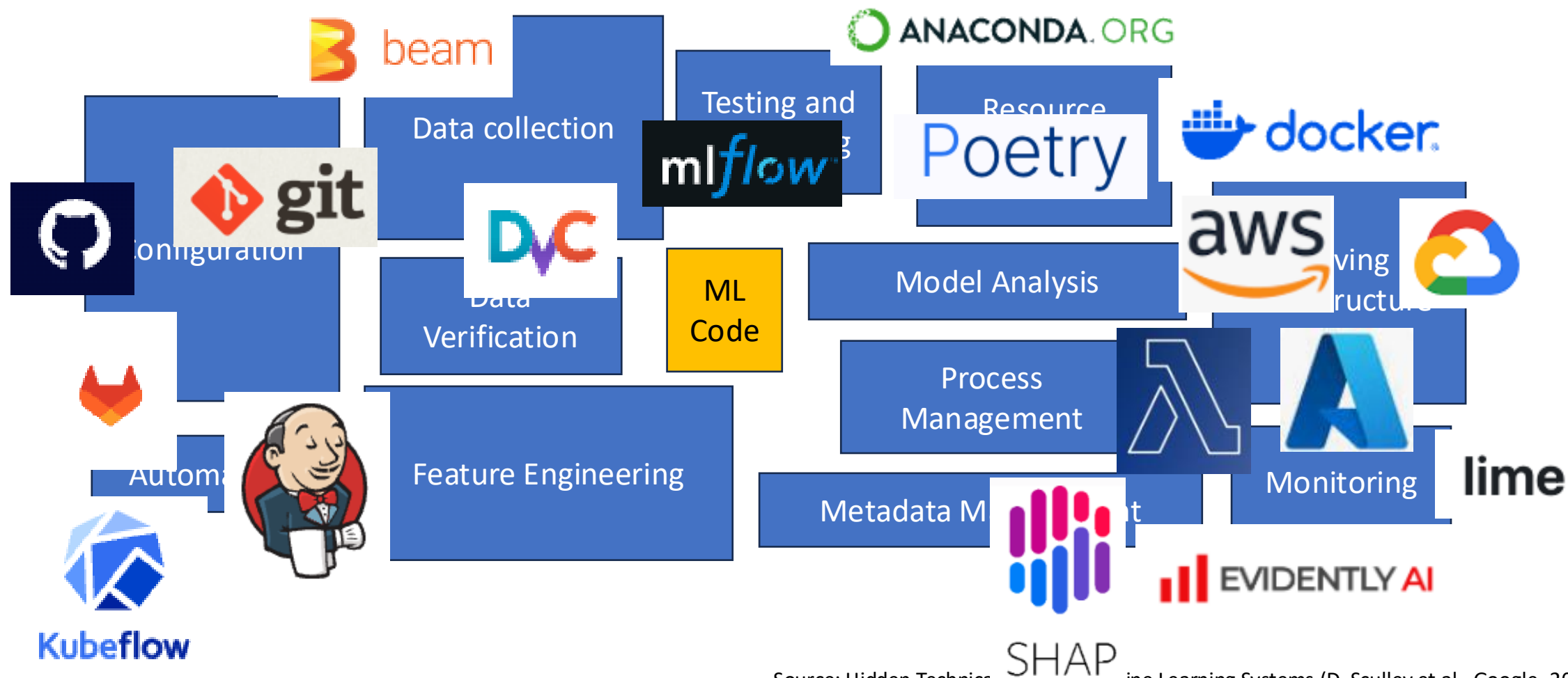
Final Thoughts

Hidden Technical Debt in ML Systems



Source: Hidden Technical Debt in Machine Learning Systems (D. Sculley et al., Google, 2015)

Hidden Technical Debt in ML Systems



Source: Hidden Technical Debt in Machine Learning Systems (D. Sculley et al., Google, 2015)

Thank You!!!